

Java SE

1. What is Java?

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2. Why Java is called as "Platform"?

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3. What are the features of Java?

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4. How java is platform independent?

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5. Which version of java have you learned? Name some of the new features added to it.

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6. What is native code?

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7. What is the difference between JDK and JRE?

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8. What is JVM? What it does?

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9. Why JVM is called as "virtual machine"?

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10. What is the lifetime of Java Virtual Machine?

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11. What are the main components of JVM? Explain them. Or Explain JVM Architecture.

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12. What is the job done by classloader?

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13. Explain the hierarchy of classloaders in java.

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14. What is the role played by Bytecode Verifier?

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15. What are the memory areas allocated by JVM?

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16. What is the difference between Java compiler (javac) and JIT ?

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17. Is JRE different for different Platforms?

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18. Difference between C++ and java in terms of object creation.

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19. What is garbage collection?

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20. Can we force garbage collection programmatically?

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21. What is an alternative for finalized method?

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22. What is the difference between `System.gc()` and `Runtime.getRuntime().gc()` ?

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24. When do you get `NullPointerException`?

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25. What is the signature of main function in java?

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26. Who invokes main() function ?

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27. Can we define more than one public class in a java source code? What is the rule of public class and file name?

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28. Difference between path and classpath.

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29. What is the rule for local member in java?

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30. What is .class file known as?

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31. How java is strong object oriented?

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32. What is finalized() method ?

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33. Why there is no sizeof operator in java?

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34. What kinds of programs u can develop using Java?

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35. Difference between C++ pointer and Java reference.

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36. You have reference type as a member of class. What is the default value it gets?

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37. What are the expressions allowed in switch block of java?

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38. In a java source code there are 3 classes define e.g class A{} class B{} and class C{}. When u compile the source code how many .class files u get and what are their names?

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39. What is java's old name?

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40. What are the technologies come under "Distributed Object System"?

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41. What do you mean by Java Enabled Browser, Web Server and Application Server?

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42. What is "this"? What is its importance?

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43. What is UnsupportedClassVersion error?

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44. What is Unicode Character Set in Java?

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45. Explain Java Execution Flow.

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46. What is the difference between Lazy and Eager Resolution?

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47. Does Java pass by reference or by value?

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48. What is instanceof operator? give an example.

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49. What is upcasting and downcasting with example?

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50. What is the risk involved in downcasting? How instanceof operator provides solution for that?

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51. Given, in C++,

MyClass m1;

MyClass m2=m1;

And in Java,

MyClass m1=new MyClass(); MyClass

m2=m1;

What is the difference?

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52. In Java copy constructor is not provided implicitly. Can we provide it explicitly and if yes when exactly it will be called?

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53. How many types of inheritance allowed in java?

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54. Why "Multiple Inheritance" is not supported in Java?

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55. Explain accessibility modifiers in java.

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56. What is super keyword?

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57. What is @Override? What does it do?

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58. What are Overriding rules in java?

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59. What is Object class?

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60. What is "ClassCastException"? What is the solution for that?

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61. Where can we apply final keyword in java?

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62. Can a class be defined a both final and abstract?

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63. Explain "System.out.println".

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64. Explain “public static void main”.

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65. Can we use “this” with static member?

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66. Explain static block or static initializer.

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67. Where have you seen an application of static block or initializer in java?

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68. What is non-static block?

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69. Can static member function access non-static data?

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70. Can non-static member function access static data?

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71. Explain constructor invocation in case of inheritance.

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72. In case of inheritance, if parent class does not have no-arg constructor, what happens? How do u solve the problem?

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73. In case of an array which variable is created automatically? Can u modify it? If no, why?

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74. What is class Class in java?

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75. What is the use of class "Class" in java?

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76. What does "forName()" function do ?

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77. When does class get loaded?

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78. What is bridge method in java?

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79. Why compiler synthesizes bridge method in case of overriding?

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80. What are the scenarios in which we have early binding in java?

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81. Explain abstract class in Java.

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82. Explain interface in Java8.

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83. Why do we need interface?

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84. Using interface reference we can invoke only those methods which are present in it.

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85. Given: interface myinterface

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```
{
void disp();
}
public class MyClass
{
public static void main(String[] args)
{
    myinterface m1,m2=null;
    m1.disp();
    m1.fun();
    m1.toString();
    m1.equals(m2);
    m1.hashCode();
}
}
```

Which of the above invocations are illegal at compile-time?

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85. When to use interface and when to use abstract class? Explain with example.

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86. How many ways we can achieve Polymorphism in Java?

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87. What is the difference between abstract class and interface in Java8?

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88. What are the default methods in Java8 interfaces?

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89. What is the actual use of “default” method in interface?

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90. What do you mean by “Class wins” rule in java8?

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91. Why it is not recommended to make a default method that redefines one of the methods from Object class?

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92. Why do we need static methods in interface?

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93. Explain this vs super in case of constructor invocation.

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94. What is equals and hashCode contract?

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95. What is the difference between == and equals() method ?

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96. Explain toString, equals and hashCode methods of Object class.

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97. What do you mean by immutable class?

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98. How to create immutable class in java?

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99. What are wrapper classes? What is their main application?

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100. JDK 1.5 onwards, we can pass primitives (i.e. int,char,float etc.) Directly to the method accepting "java.lang.Object". Does it mean that there is no use of Wrapper classes now?

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101. Difference we can pass primitives (i.e. int,char,float etc.) Directly to the method accepting.

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102. What is autoboxing and unboxing?

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103. Autoboxing and unboxing are limited to compiler only. What do u mean by this?

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104. What technique is used to make variable number of arguments possible in java?

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105. When you try to print any reference variable in java, u get some meaningful output. What do u mean by this?

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106. Why can't we instantiate abstract class?

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107. Can abstract class have constructors in Java?

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108. Can abstract classes implements interface in Java? Do they require to implement all methods?

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109. Can abstract class be final in Java?

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110. Can abstract class have static methods in Java?

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111. Is it necessary for abstract class to have abstract method?

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112. Can abstract class contains main method in Java?

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113. What do you mean by concrete class?

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114. What do you mean by "Functional Interface"?

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115. What Is an Exception?

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116. Why do we need Exceptions?

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117. Describe the exception hierarchy in java.

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118. What is the difference between “Exception” and “Error”?

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119. What happens when an exception is raised in a program?

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120. Why do you handle the exception?

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121. How to handle the exception?

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122. When you write one try and multiple catch, what care u should take?

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123. Can we write catch block in a following manner: catch(Object ref)

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{ }
```

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124. What is finally block? Why do we need it?

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125. In which scenarios finally block will not get executed?

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126. Define checked and unchecked exception.

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127. In java, adjectives ending in -able are interfaces Serializable, Comparable , Cloneable etc... So why is Throwable a class and not an interface?

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128. When will you create a checked exception?

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129. How do you create checked and unchecked exception?

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130. Compiler enforces u to deal with checked exception. What do u mean by this?

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131. Difference between throw and throws.

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132. What is ARM block?

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133. Is it recommended to handle "NullPointerException", "ClassCastException" etc. using java code?

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134. What is Re-throwing an exception in java?

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135. What are the legal combinations of try, catch and finally blocks?

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136. What are some of the best practices of Exception Handling?

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137. When deciding on checked exceptions vs. unchecked exceptions, ask yourself, "What action can the client code take when the exception occurs?"

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138. What is the difference between `NoClassDefFoundError` and `ClassNotFoundException`?

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139. What are the performance implications about exception handling?

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140. How do you convert checked exception to unchecked exception?

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141. What is process?

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142. What is thread?

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143. What is meaning of Process-based multitasking and Thread-based multitasking? Give examples.

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144. What do you mean by context switching?

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145. Difference between Process-based multitasking & Thread-based multitasking.

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146. What is one of the main application of multi-threading in java? Explain.

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147. What are the main steps in a typical multithreading application?

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148. Can we directly call run() method ? What happens in that case?

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149. What is the default priority every user thread gets?

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150. Explain "extends Thread".

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151. Explain “implements Runnable”.

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152. In extends Thread, if we don't override “run()” method what will happen ?

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153. In implements Runnable, if we don't define “run()” method what will happen ?

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154. What is the Difference between "implements Runnable" and "extends Thread" in Java?

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155. You are extending Thread or Implementing Runnable. You need to deal with some kind of checked exception inside run() method. What will u do? You will declare it or handle it inside run() ? Give reason.

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156. Normally when does thread die?

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157. Can we restart dead thread?

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158. Why start() method is not final?

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159. What is main thread?

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160. Explain Race condition with example.

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161. How do we avoid Race condition?

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162. What is synchronization?

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163. What exactly happens when we use synchronized keyword?

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164. Difference between synchronized method and block.

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165. Once a thread acquires the lock on an object, what are the things it has control on?

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166. What do you mean by “Thread-Safe” class?

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167. What is the drawback of synchronization?

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168. What is class lock?

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169. What is the need for Threads Communication?

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170. What are the methods used for Thread communication? Explain them in brief.

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171. If no thread is in the wait pool, and we invoke notify or notifyAll, what will happen?

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172. wait(), notify() and notifyAll() methods can be called only from synchronized method or a block. Why it is so?

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173. Difference between sleep and wait.

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174. What do you mean by Daemon Thread? Give an example.

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175. What is InterruptedException?

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176. What do you mean by Thread's born state?

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177. What is the difference bet'n runnable and running state?

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178. From blocked state, thread goes to runnable or running state?

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179. You are using "implements Runnable" technique for multithreading application. What happens when u pass an object implements Runnable to the Thread class constructor and what happens if you don't pass such object to Thread class constructor?

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180. What is Re-Entrant lock in java?

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181. What is the Difference between ReentrantLock and implicit lock?

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182. What is the drawback of Reentrant lock?

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183. What is lock fairness in case of Reentrant lock?

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184. What is the difference between “tryLock()” and “tryLock(long time, TimeUnit wait)”?

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185. Explain "Condition Objects" in case of Reentrant lock.

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186. What are Executors?

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187. What do you mean by thread pooling?

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188. How does Executor work?

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189. What is the advantage of Thread pool over creating threads from the scratch?

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190. What is ForkAndJoin framework?

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191. What is Work-Stealing algorithm in Java?

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192. What is the difference between Fork/Join Framework and ExecutorService?

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193. Name the existing Implementations in JDK which use Fork/Join Framework.

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194. What do you mean by serialization and deserialization?

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195. What is the need of serialization?

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196. Object serialization was added to java in order to support which two features?

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197. What are the requirements we need to meet in order to serialize an object?

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198. What is the difference between Serializable and Externalizable?

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199. What is serialVersionUID in case of Serialization?

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200. What happens when we deserialize an object?

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201. What is transient variable in Java?

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202. When should we use transient keyword in java?

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203. What are the things get written when u serialize an object?

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204. If a class is Serializable but its super class is not, what will be the state of the instance variables inherited from super class after deserialization?

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205. Suppose super class of a new class implements Serializable interface, how can you avoid new class to be serialized?

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206. Suppose you have a class which you serialized it and stored in persistence and later modified that class to add a new field. What will happen if you deserialize the object already serialized?

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207. Which kind of variables is not serialized during Java Serialization?

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208. Does “java.lang.Object” class implement “Serializable”? Explain with reason.

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209. What happens when one of the members in the class doesn't implement Serializable interface?

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210. What happens when base class doesn't implement Serializable interface but sub class implements?

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211. Can you Customize Serialization process or can you override default Serialization process in Java?

212. While customizing serialization process, if we define “readObject()” and “writeObject()” with the accessibility modifiers other than “private” what will happen?

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213. Can we transfer a Serialized object via network?

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214. How many ways we can create objects in java?

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215. What is the difference between Standard IO and NIO?

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216. What are Generics in Java?

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217. How does Generic provide Type-safety?

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218. What do you mean by type-erasure?

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219. Explain <? extends T> in java generics.

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220. Explain <? super T> in java generics.

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221. What is the rule of polymorphism and generics?

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222. Why the following statement will not work?

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223. Given the following two scenarios:

```
a) public void addAnimal(Animal arr[])  
{  
    System.out.println(arr[0]);  
}
```

```
Dog d[]={new Dog(),new Dog()}; addAnimal(d);
```

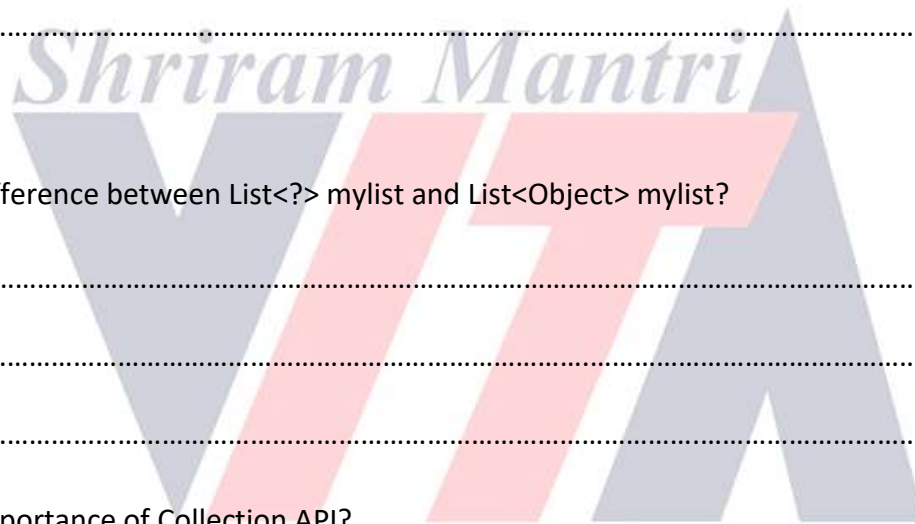
```
b) public void  
   addAnimal(List<Animal> ref)  
   {  
       System.out.println(ref);  
   }  
   addAnimal(new ArrayList<Dog>());
```

Explain with reason which scenario is legal at compile-time and which is not.

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224. What is the difference between `List<?> mylist` and `List<Object> mylist`?

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225. What is the importance of Collection API?

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226. What are the basic interfaces of Java Collections Framework?

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227. Why Collection doesn't extend Cloneable and Serializable interfaces?

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228. Why Map interface doesn't extend Collection interface?

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229. What is an Iterator?

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230. What is difference between Enumeration and Iterator interface?

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231. What is the difference between Iterator and ListIterator?

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232. What are different ways to iterate over a list?

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233. What do you understand by iterator fail-fast property?

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234. What is difference between iterator being fail-fast and fail-safe?

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235. Why there are no concrete implementations of Iterator interface?

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236. How HashMap works in Java?

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237. What are the changes java8 brought to the Map implementations?

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238. Can we use any class as Map key?

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239. How to convert an array of String to List?

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240. How HashSet store elements?

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241. Why we use Map interface? What are main classes implementing Map interface?

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242. Difference between HashMap and TreeMap?

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243. Difference between Set and List.

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244. Difference between HashMap and Hashtable

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245. Difference between Vector and ArrayList.

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246. Difference between Iterator and Enumeration.

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247. Difference between ArrayList and LinkedList.

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248. How to make a collection read only?

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249. How to make a collection thread safe?

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250. Why there is no method like Iterator.add() to add elements to the collection?

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251. What do you understand by iterator is fail-safe?

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252. How to avoid ConcurrentModificationException while iterating a collection?

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253. What is UnsupportedOperationException?

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254. What is the importance of hashCode() and equals() methods? How they are used in Java?

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255. What is the difference between CopyOnWriteArrayList and ArrayList in Java?

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256. Which implementation of the List interface provides for the fastest insertion of a new element into the middle of the list?

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257. How do you remove an entry from a Collection? and subsequently what is difference between remove() method of Collection and remove() method of Iterator, which one you will use, while removing elements during iteration?

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258. What is difference between Synchronized Collection and Concurrent Collection?

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259. How does HashSet is implemented in Java, How does it uses Hashing?

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When do you use ConcurrentHashMap in Java?

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260. Mention some of the Best Practices of Collection API.

261. What is hash collision? How does map implementations deal with it?

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262. What is the difference between “put()” method of Map vs “add()” method of Set?

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263. How does “put()” method of map actually works?

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264. What if we add the another value object inside map with same key (with matching hashCode and equals) as entered before? Or How do map implementations ensure the uniqueness of keys?

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265. Why ListIterator has add() method but Iterator doesn't or Why add() method is declared in ListIterator and not on Iterator.

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266. When does ConcurrentModificationException occur on iteration?

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267. What is the difference between comparator and comparable in java?

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268. What is CopyOnWriteArrayList?

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269. Explain ConcurrentHashMap in java.

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270. How will you retrieve Value object from map implementations if two Keys will have same hashCode?

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271. Draw the picture of entry table in case of map implementations.

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272. What happens On HashMap in Java if the size of the HashMap exceeds a given threshold defined by load factor?

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273. What do you mean by rehashing in case of map implementations?

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274. Why String, Integer and other wrapper classes are considered good keys?

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275. How does map resolves collision?

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276. How do you traverse through a collection using its Iterator?

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277. What are the advantages of ArrayList over arrays?

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278. Why insertion and deletion in ArrayList is slow compared to LinkedList?

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279. Why are Iterators returned by ArrayList called Fail Fast?

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280. How do you decide when to use ArrayList and When to use LinkedList?

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281. Difference between ArrayList and Vector.

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282. Explain Collection API hierarchy.

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283. What is the difference between Enumerator, Iterator and ListIterator?

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284. What is the difference between Hashtable and ConcurrentHashMap?

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285. What is the difference between Synchronized Collection classes and Concurrent Collection Classes?

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286. What is the drawback of synchronized collections classes, Hashtable and Vector, and the synchronized wrapper classes, Collections.synchronizedMap and Collections.synchronizedList etc.?

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287. What is heap pollution?

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288. Even after creating Collections.synchronizedList(new ArrayList()), why do we need to use "synchronized" block?

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289. Discuss the problem of thread-safety with synchronized collection classes as well as synchronized wrapper classes.

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290. What is the requirement of putting any key inside TreeMap or element inside TreeSet?

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291. What is the difference between C++ class template and Java Generics?

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292. Explain HashMap changes in Java8.

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293. How will you make Collections readOnly?

294. Suppose there is an Employee class. We add Employee class objects to the ArrayList. Mention the steps need to be taken, if I want to sort the objects in ArrayList using the employeeid attribute present in Employee class.

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295. How to prevent Singleton Pattern from Reflection, Serialization and Cloning?

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296. In how many ways you can create string objects in java?

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297. What is string pool?

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298. What is special about string objects as compared to objects of other derived types?

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299. What do you mean by mutable and immutable objects?

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300. Why StringBuffer and StringBuilder classes are introduced in java when there already exist String class to represent the set of characters?

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301. How many objects will be created in the following code and where they will be stored in the memory?

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302. How do you create mutable string objects?

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303. Which one will you prefer among “==” and equals() method to compare two string objects?

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304. Which class will you recommend among String, StringBuffer and StringBuilder classes if I want mutable and thread safe objects?

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305. Where exactly string constant pool is located in the memory?

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306. Why You Need String Constant Pool?

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307. What Is String Intern?

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308. What do you mean by 'String Literals Are Automatically Interned'?

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309. What is the use of interning the string?

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310. Why String is Immutable or Final in Java?

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311. What is the similarity and difference between String and StringBuffer class?

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312. What is the similarity and difference between StringBuffer and StringBuilder class?

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